Assignment Data Vault

Please note that the examples given here are just conceptual examples, don't use them for syntax, naming convention or whatsoever.

They are provided just to help you on your way trough the steps;

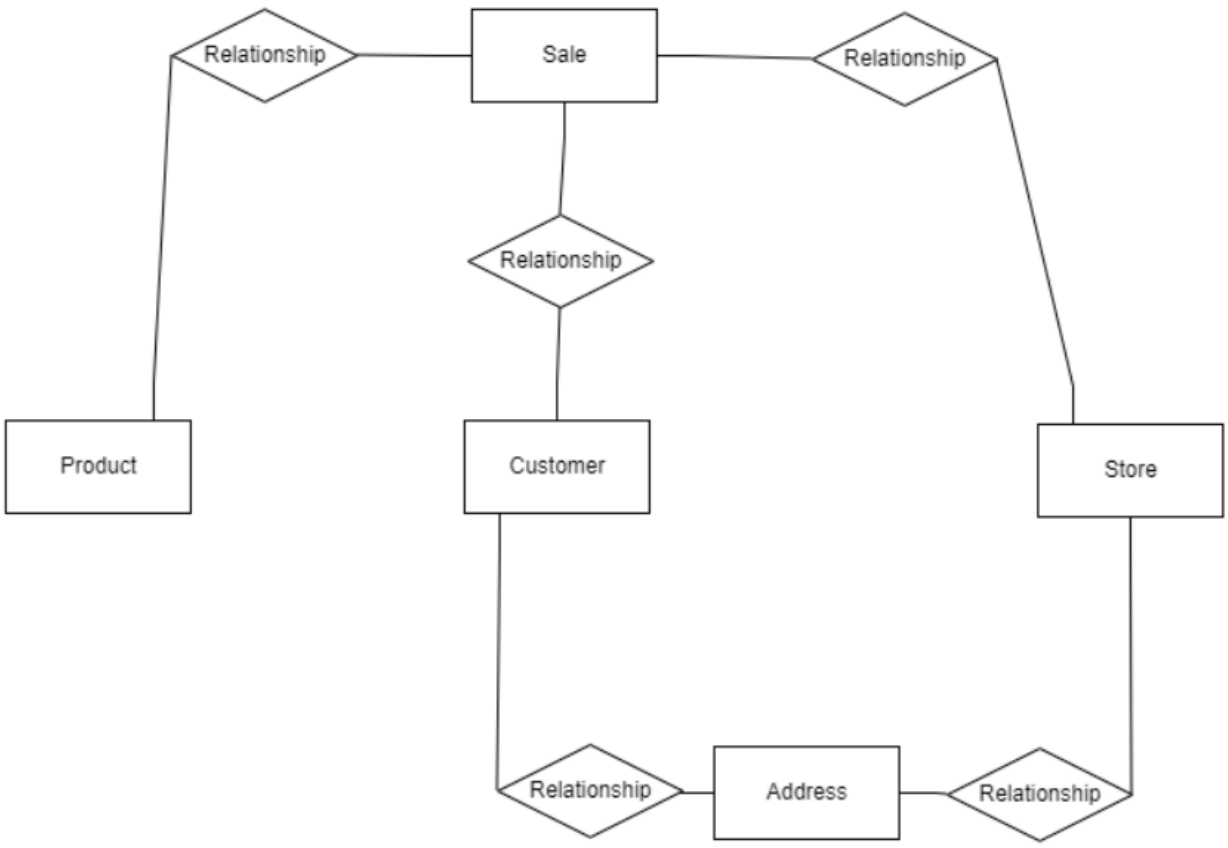
1 Analyze the Source Files

For each source file, identify the logical concepts in the data, and visualize them in a conceptual model

Source Files:

- ACI_ASQ_SURVEY_MAIN
- FILE_AIRPORT_PANEL
- FILE_QUESTION_RESPONSIBLE
- REF_ASQ_AIRPORT_SIZE

For example:



1.1 Solution

We have four source files:

- file_asq_survey
- file_question_responsible
- file_airport_panel
- map_asq_airport_size

1.1.1 ASQ_SURVEY file

Main Business Key

In this file we have a lot of information.

The main entity is of course the survey itself, that is the Questionnaire, the paper where the interviewer registered the answers of a passenger in an airport.

That survey is identified by the column "QuestNo" (column BP in Excel) almost at the end of the file.

So, **SURVEY** is definitely a candidate for a hub.

What else?

Airport

This is also an important business key, since we (Thomas More Airport) are an airport and want to compare our scores against those of other airports.

In other source files (see bellow) we have also a list of airports to be compared against, and also airport sizes.

AIRPORT is also a candidate for a Hub.

Airport size and region are descriptive fields in our context and will be stored in a satellite.

Flight

In the context of surveys, the flight information (flight number, airline and date / time) is not of any importance, it's just additional info, that we won't even use.

So, no hub for this in our survey context.

All this data is supposed to be stored in a satellite.

Question (non-score)

Questions at the other hand, is very important, especially the Scored Questions.

But the informational-questions are just additional info to some topics of the survey, they will not be used for the core of the requirements, so no hub for there non-scored questions.

Therefore, they will be stored in a satellite.

Scored Questions

The scored questions at the other hand, is very important, they make the core of the requirements. These scored questions are also related to a responsible, but that is for the other flat file later down.

These scored **QUESTIONS** will be used to define a hub

Personal Info

The personal information found in the survey file will be used for some reporting, but this personal info is part of a survey, each sheet is answered by a person, so this will not be a candidate for a hub. It's more like descriptive data of the survey, and for this reason will be hosted by a satellite.

Others

The other info found in the Survey file (like Terminal, Weight Factor, Gate Number ...) are in our context more descriptive information than core elements.

So no hubs here, all info goes intoo a satellite.

But, imagine that we also have an application for monitoring the terminal and gate occupancy and security, then they would be candidates for being hubs, allowing us also to interconnect the Occupancy Monitoring app data to the survey data.

Scores

The scores are also core to our business requirements.

But you must see the score like a measure. And this measure is related to different business elements:

- survey
- question
- airport

A score is kind of descriptive to a combination of business keys and will be hosted by a satellite.

The combination (join) of the three business keys are represented by a link table.

This means that we will have a SAT on a LINK table for the scores.

1.1.2 QUESTION_RESPONSIBLE file

In this file you find two major entities:

- Question (and his classification)
- Responsible

Question

In this list of questions, you'll notice that only the scored-questions are listed. All the other questions from the survey are not scored, and thus not belonging to this business entity.

As already specified in the previous chapter, the scored **QUESTION** is definitely a candidate for being a hub here.

The hierarchy itself, where a question belongs to, is considered as descriptive info.

Responsible

The **RESPONSIBLE** is also a candidate for being a hub. In a bigger picture where we have an HR source application, the responsible could be linked to employee data for example.

1.1.3 AIRPORT_PANEL

The airport panel file lists a couple of airports we want to compare ourselves to.

A stated before, **AIRPORT** is one of the core elements here, and will be represented by a hub.

1.1.4 AIRPORT_SIZE

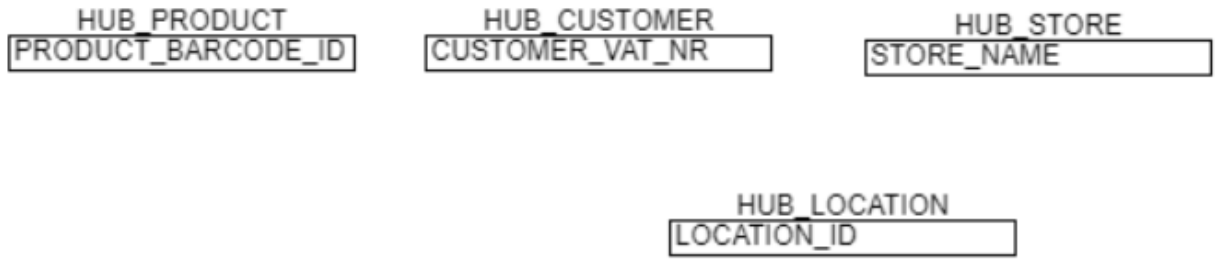
This file contains descriptive data of a descriptive attribute of an airport, with historization.

The words “descriptive" and "history" are hints to make you think about the definition of a satellite.

2 Define the entities in the files that you think are candidates for being a HUB

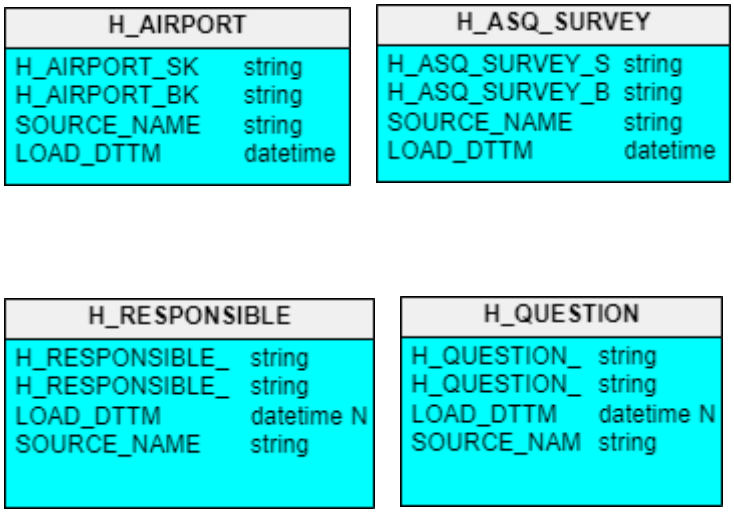
Identify the HUBs and the Business_Keys that are defining them.

For example:



2.1 Solution

We have identified 4 HUB's in the previous section:



Remember that in a HUB, you have only 4 columns:

- the hash of the BK, like H_ASQ_SURVEY_SK
- the business key itself, like H_ASQ_SURVEY_BK
- the source name
- the load datetime

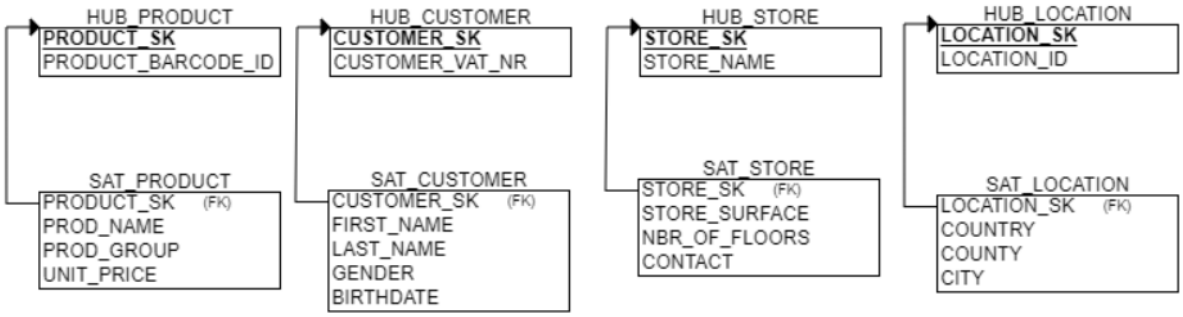
The column names are just column names, but try to define a standard, for example:

- Table Name:
    - prefix: H_
    - business key
- Columns:
    - hask key: tablename + _SK (surrogate key)
    - business key: tablename + _BK (business key)
    - source_name or src_name ...
    - load_dttm or load_time ...

3 For each HUB, define the SATELLITE you think are necessary

The attributes that you need for reporting go in the SATs.

For example:



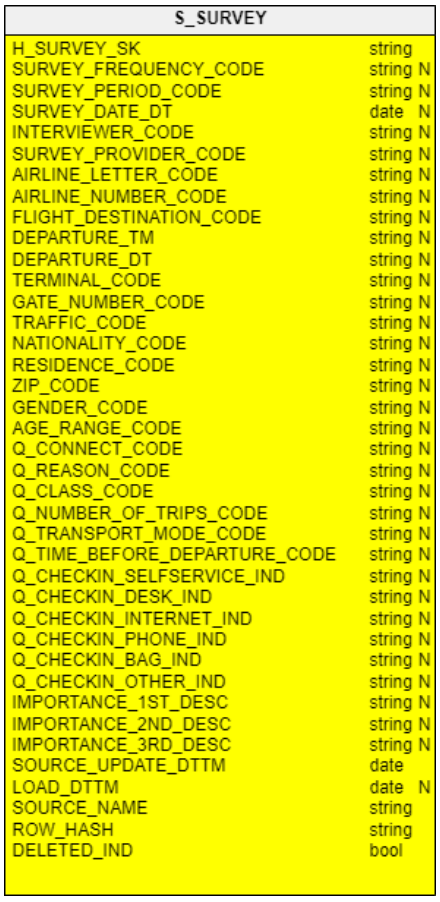
3.1 Solution

All the descriptive fields of a business key will be hosted by one or more satellites (you can split them by business context in multiple satellites if you want)

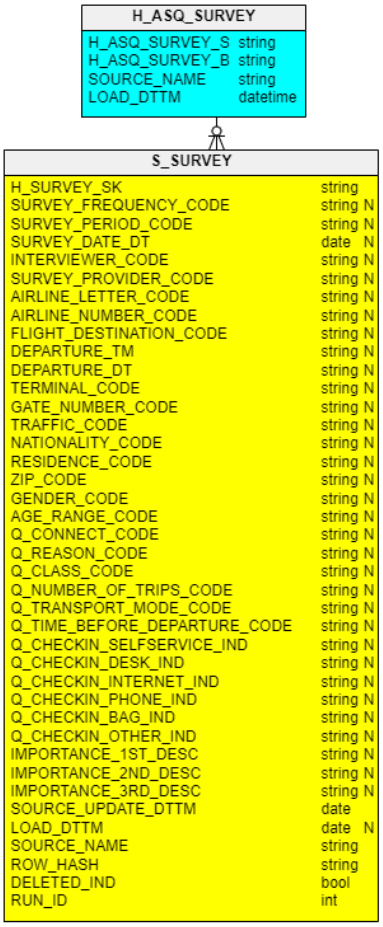
Remember that the satellite always contains:

- the business key (hash) it is related to, like H_AIRPORT_SK
- all the descriptive fields you want to put in the satellite (without any soft business rule applied to it!!!), like AIRPORT_SIZE (this are meaningless codes from 1 to 6) or AIRPORT_REGION_CODE
- a ROW_HASH, being a hash of all the descriptive attributes
- Load date
- source name

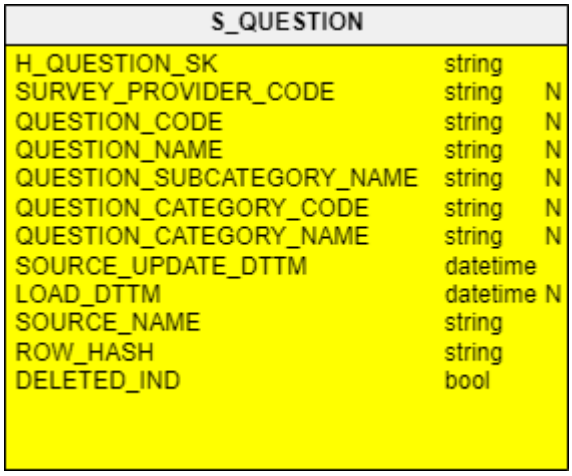
3.1.1 Survey



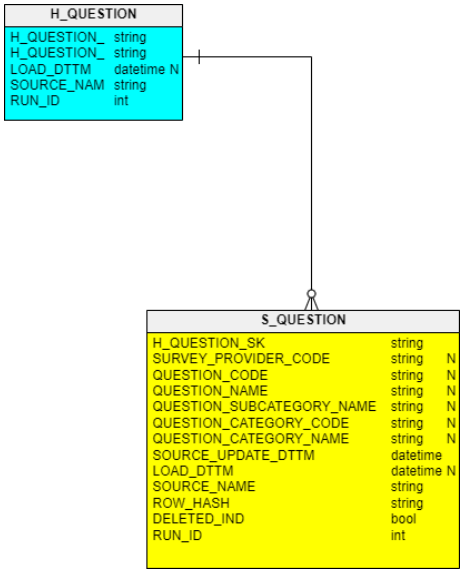
and linked to its HUB:



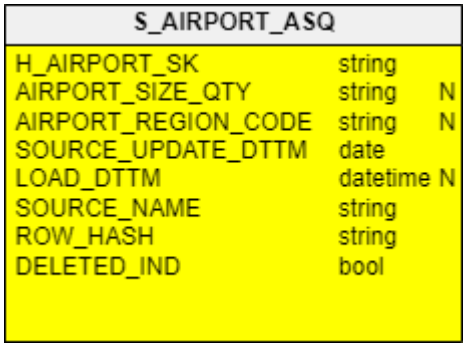
3.1.2 Question



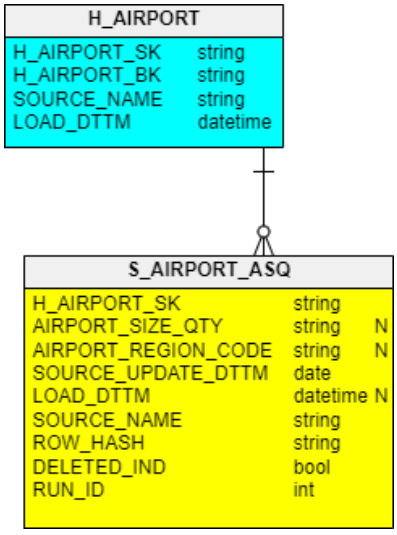
and linked to its HUB:



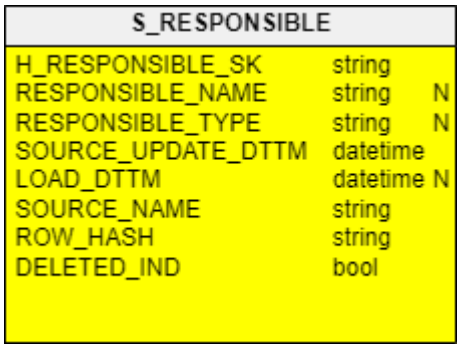
3.1.3 Airport



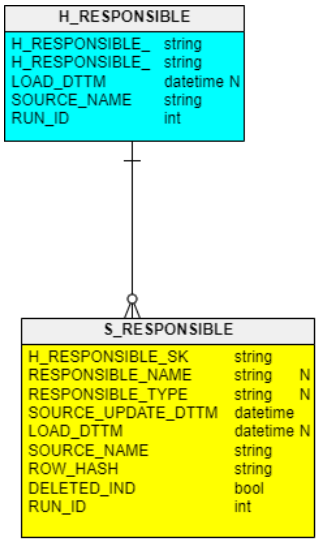
and linked to its HUB:



3.1.4 Responsible



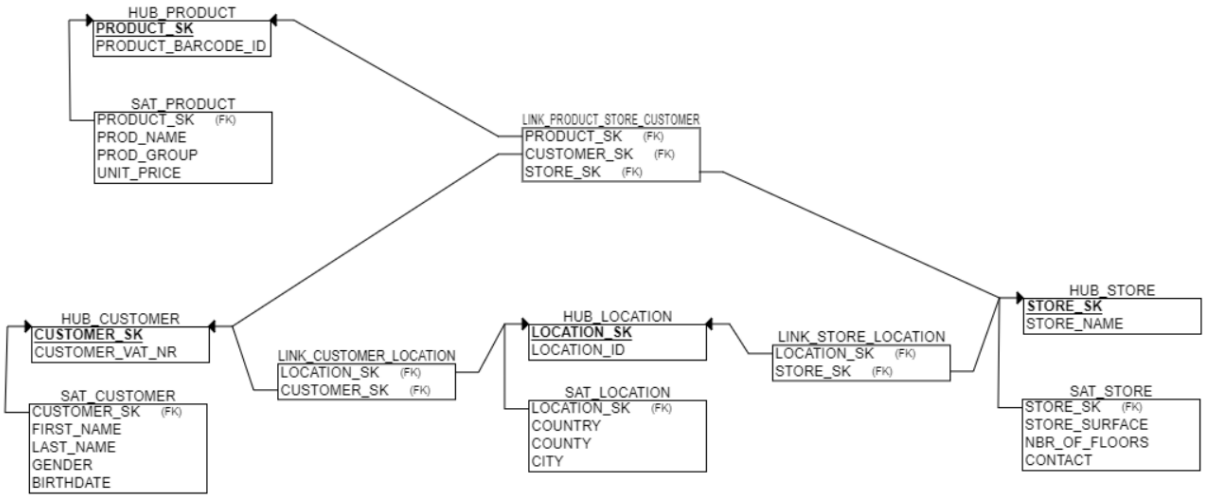
and linked to its HUB:



4 Figure out what is linked together and how

Remember, LINKs can represent joins, but also events or transactions

For example:



4.1 Solution

In the source files, you probably found that:

- a question is linked to a responsible
- a question is also linked to a survey
- an airport is linked to a survey

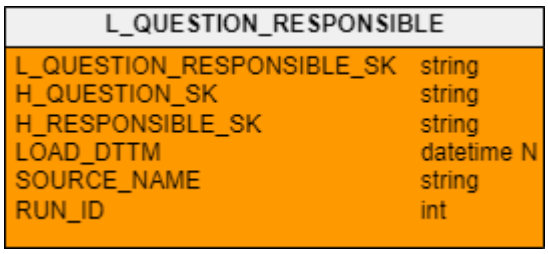
So we have to sets of joins here:

- question - responsible (representing an association)
- airport - survey - question (defining also the granularity of a score)

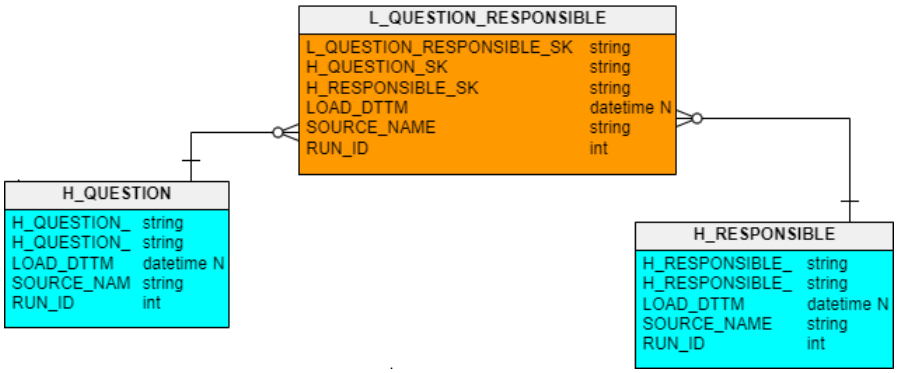
Remember that in a link table we only have the following columns:

- the hash of all the business key hashes
- all the business key hash columns of the linked hubs
- load datetime
- source name
- (in DV2.0, the business keys from the hub's are also added)

4.1.1 Link Question - Responsible



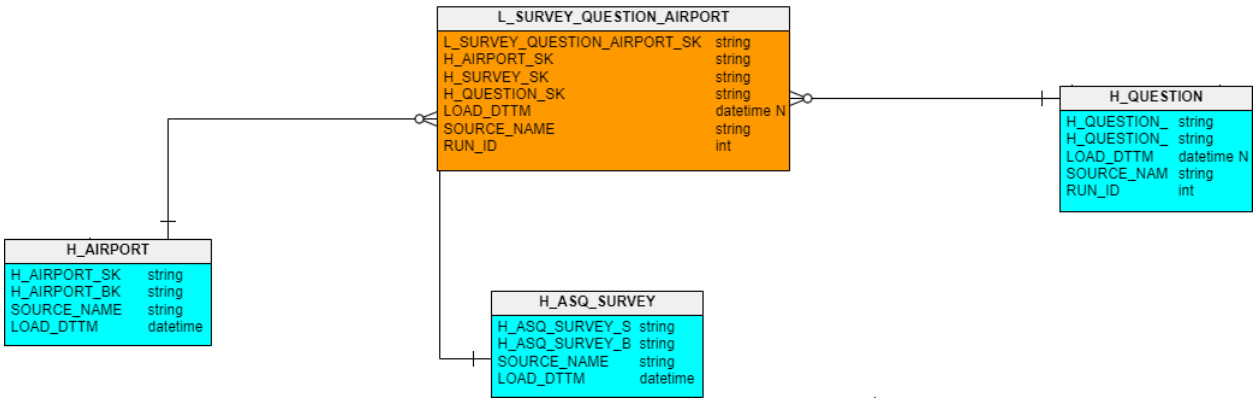
and linking the two hubs:



4.1.2 Link Survey - Question - Airport

L_SURVEY_QUESTION_AIRPORT	
L_SURVEY_QUESTION_AIRPORT_SK	string
H_AIRPORT_SK	string
H_SURVEY_SK	string
H_QUESTION_SK	string
LOAD_DTTM	datetime N
SOURCE_NAME	string
RUN_ID	int

linking the three hubs together:

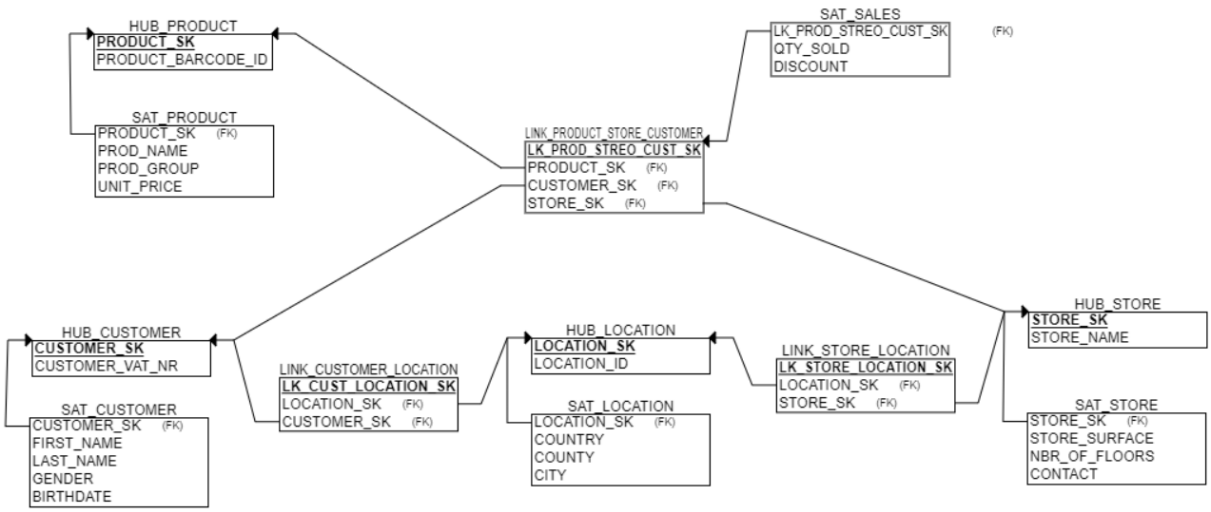


5 What about the measures?

You need also your measures for the reporting.

Where are you storing them, and how to integrate the in the DV model?

For example:



5.1 Solution

The measures that we can define in the source file are:

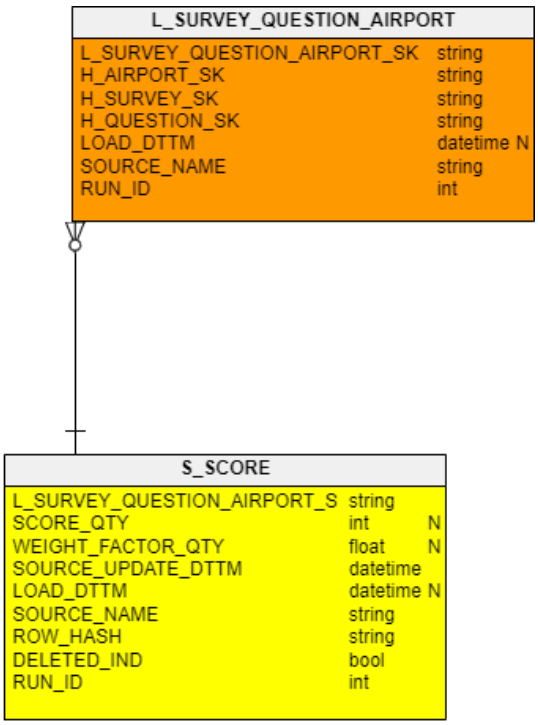
- the score (0-5)
- the weight factor (somewhere between 1.05 and 0.95)

We are NOT going to calculate the weighted score here in the data vault, since NO soft business rules, or any changes in the data are allowed.

S_SCORE		
L_SURVEY_QUESTION_AIRPORT_S	string	
SCORE_QTY	int	N
WEIGHT_FACTOR_QTY	float	N
SOURCE_UPDATE_DTTM	datetime	
LOAD_DTTM	datetime	N
SOURCE_NAME	string	
ROW_HASH	string	
DELETED_IND	bool	
RUN_ID	int	

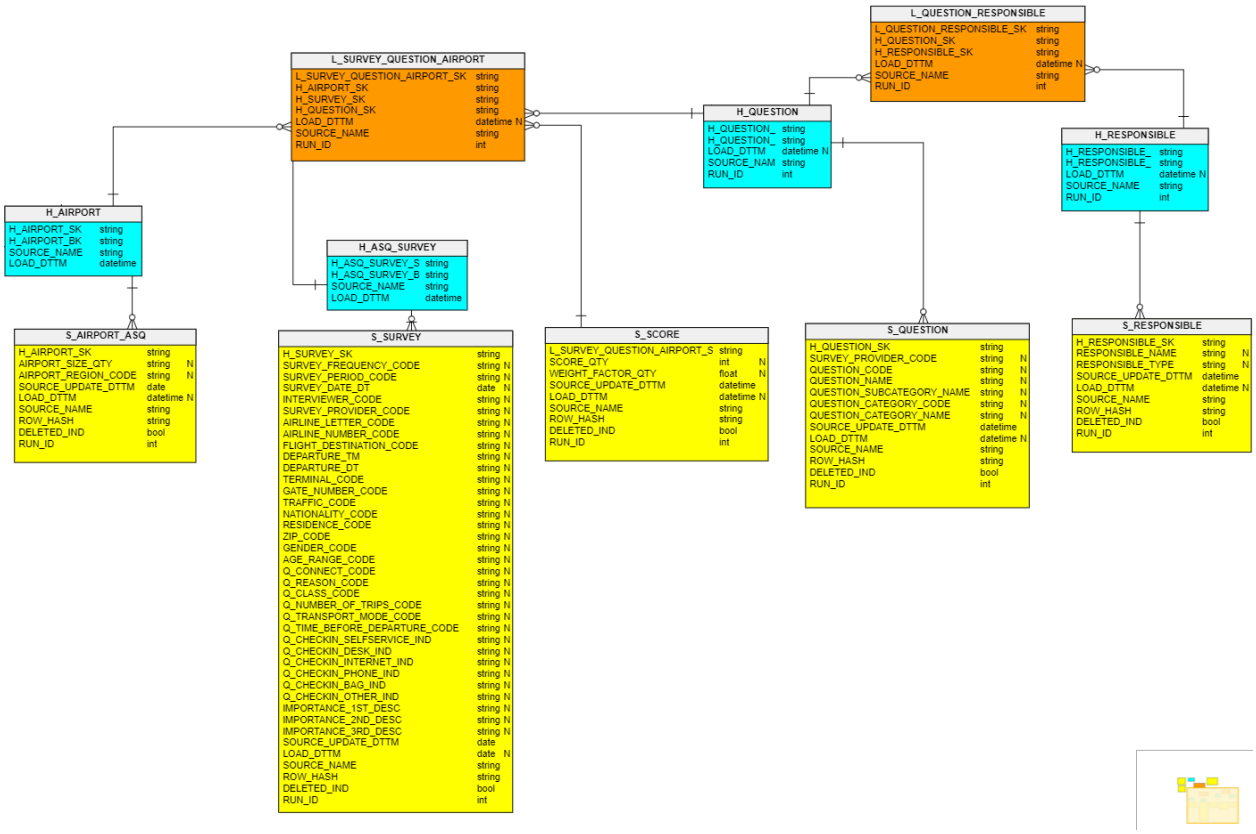
The difference with the other satellites we already defined previously, is that this satellite is linked to a LINK table.

You see the first column; it refers to the HASH key of the link table.



6 Solution: All steps grouped together

The Dava Vault will look like follows, the HUB's connected via the LINK's and the SAT's hosting the descriptive data of the HUB's:



In the specific case of this surveys, one important step to model the resulting Data Vault, is pivoting the ASQ_SURVEY file.

6.1 Pivot the Survey data

The issue with the provided flat file is that all the scored questions are provided as separate columns.

This will not work for the reporting; imagine you have a chart with a filter to choose a specific question. If each question is defined in a separate column, you cannot filter on it, you will have to replace a question by another, instead choosing a value from a "list of values".

So, all the scored question columns have to be pivoted into one scored-question column:

survey	qst1	qst2	qst3	qst4
id1	2	1	0	5
id2	3	5	2	4
id3	5	4	3	5
id4	4	2	2	3



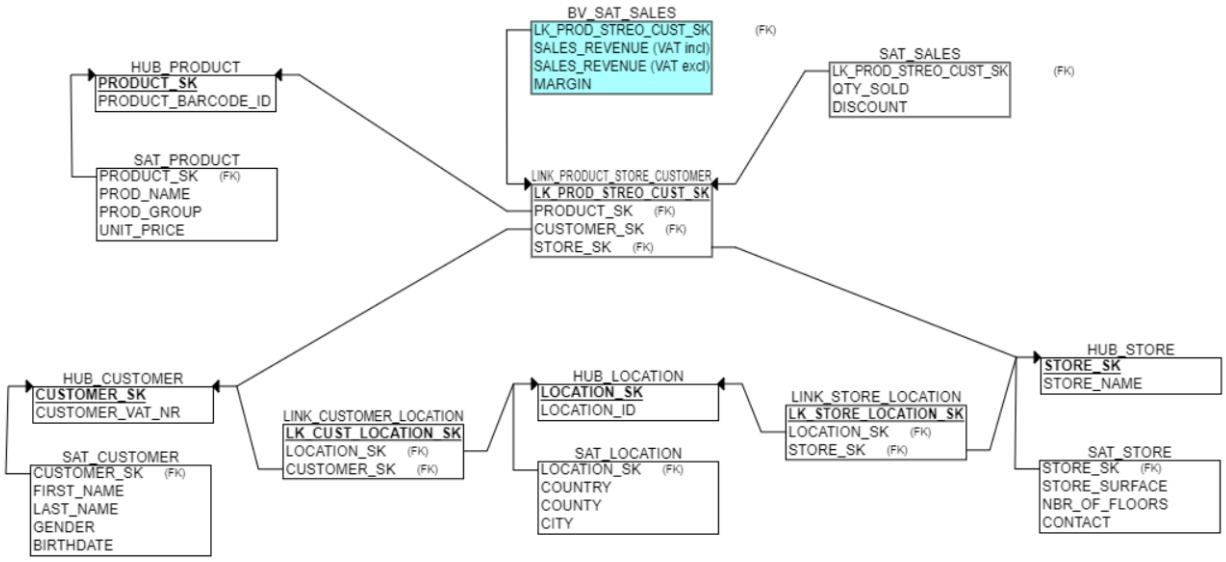
survey	question	score
id1	qst1	2
id2	qst1	3
id3	qst1	5
id4	qst1	4
id1	qst2	1
id2	qst2	5
id3	qst2	4
id4	qst2	2
id1	qst3	0
id2	qst3	2
id3	qst3	3
id4	qst3	2
id1	qst4	5
id2	qst4	4
id3	qst4	5
id4	qst4	3

7 Find the candidates for a Business Vault

Since we don't apply soft business rules to the Data Vault, find the calculations you would want to pre-calculate in the appropriate table(s)

They will be used in the reporting layer (datamart) , and by that way avoid a calculation in each datamart with the risk of different rules applied.

For example:



7.1 Solution

A couple of Business Vault tables can be defined it this context.

As you probably noticed, till here there is no trace of the AIRPORT_PANEL nor the SIZE_DESCRIPTION and RANK and POSITION calculation.

Also, at this point, in the data vault, we have all surveys. Remember that every paper-sheet has its unique ID. All this data is loaded in the Data Vault. But nobody is interested in this level of detail. So we will aggregate the surveys at an higher level in the Business Vault.

7.1.1 Airport

For the airport business entity, we have two missing attributes till now:

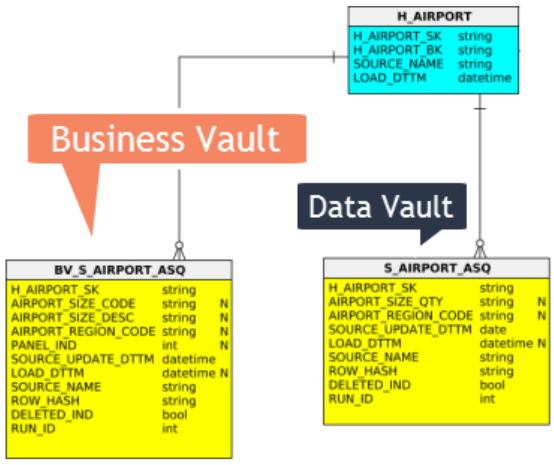
- airport panel
- airport size (is an ID now, must be translated in something usable)

Since the level of granularity of the airport doesn't change, we will keep the existing HUB (H_AIRPORT).

But the additional info will be added in a BUSINESS VAULT SATELLITE:

BV_S_AIRPORT_ASQ		
H_AIRPORT_SK	string	
AIRPORT_SIZE_CODE	string	N
AIRPORT_SIZE_DESC	string	N
AIRPORT_REGION_CODE	string	N
PANEL_IND	int	N
SOURCE_UPDATE_DTTM	datetime	
LOAD_DTTM	datetime	N
SOURCE_NAME	string	
ROW_HASH	string	
DELETED_IND	bool	
RUN_ID	int	

So the resulting AIRPORT domain will look like this:



7.1.2 Survey

Since the source data is far to detailed for our business requirements, we will have to create a new SURVEY HUB and LINK table in the Business Vault.

This survey ID is of no use here, so we will define a new entity for survey: SURVEY_SPECIFICATION.

In business key of a survey specification will be:

- the survey provider
- the survey period (date like 202010for monthly surveys, 2020Q4 for quarterly surveys)

This will result in a Business Vault Hub for Survey Specifications like:

BV_H_SURVEY_SPEC		
BV_H_SURVEY_SPEC_	string	
BV_H_SURVEY_SPEC_	string	
LOAD_DTTM	datetime	N
SOURCE_NAME	string	
RUN_ID	int	

The descriptive attributes for a survey specification will be:

- the survey provider code
- the survey period code (202010, 2020Q4...)
- the survey frequency (M or Q, useful for filtering a periodicity)
- the start and end date of each period code

The new business vault satellite for survey spec will look like:

BV_S_SURVEY_SPEC		
BV_H_SURVEY_SPEC_SK	string	
SURVEY_PROVIDER_CODE	string	N
SURVEY_PERIOD_CODE	string	N
SURVEY_FREQUENCY_CODE	string	N
SURVEY_PERIOD_START_DT	datetime	N
SURVEY_PERIOD_END_DT	datetime	N
SOURCE_UPDATE_DTTM	datetime	
LOAD_DTTM	datetime	N
SOURCE_NAME	string	
ROW_HASH	string	
DELETED_IND	bool	
RUN_ID	int	

7.1.3 New Aggregation Level

Previously, the "aggregation level" for the scores was:

- airport
- survey
- question

But this was far to much detailed for our business requirements.

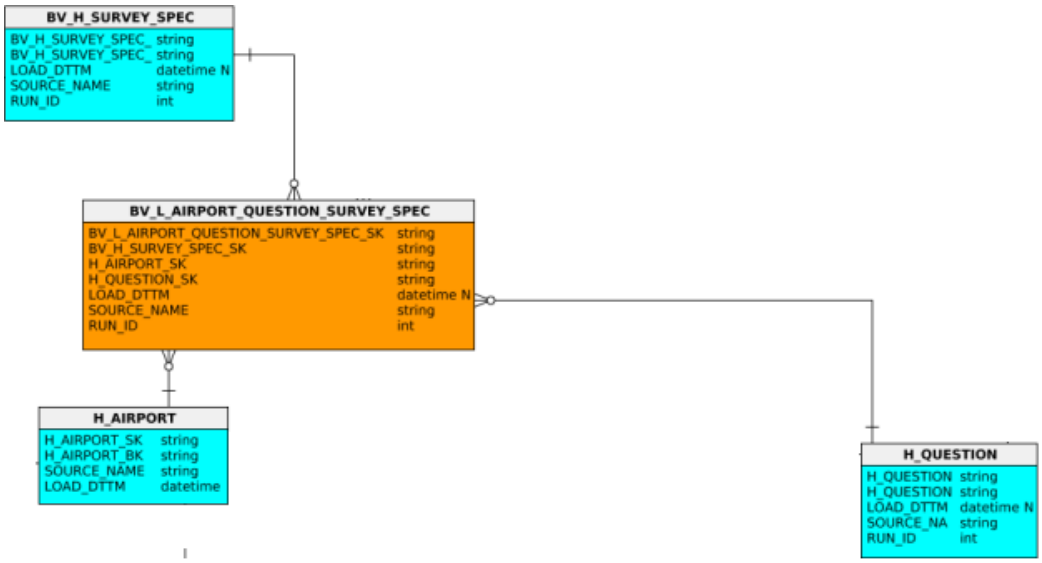
So, now that we have redefined the level of detail for the surveys to just survey specifications, we will also have change the aggregation level of the scores and other related calculations.

First step will be to create a LINK table in the business vault, linking the correct HUB's together:

- H_AIRPORT (unchanged)
- H_QUESTION (unchanged)
- BV_H_SURVEY_SPEC (new)

BV_L_AIRPORT_QUESTION_SURVEY_SPEC	
BV_L_AIRPORT_QUESTION_SURVEY_SPEC_SK	string
BV_H_SURVEY_SPEC_SK	string
H_AIRPORT_SK	string
H_QUESTION_SK	string
LOAD_DTTM	datetime N
SOURCE_NAME	string
RUN_ID	int

resulting in the following joins:



7.1.4 Calculated Measures

Where the score was at survey level in the data vault satellite S_SCORE, they will now be calculated at the aggregation level of the new business vault link table BV_L_AIRPORT_QUESTION_SURVEY_SPEC.

Additionally, to the score, we will also calculate:

- the RANK
- the POSITION
- the WEIGHTED_SCORE (sum of the scores * weight_factor)
- the COUNT of the scores

Remember, we won't calculate the average score here, because an average is always defined by the detail of its aggregation level. And if in the report someone wants to change the aggregation level, he will have to calculate averages of averages, resulting in wrong figures.

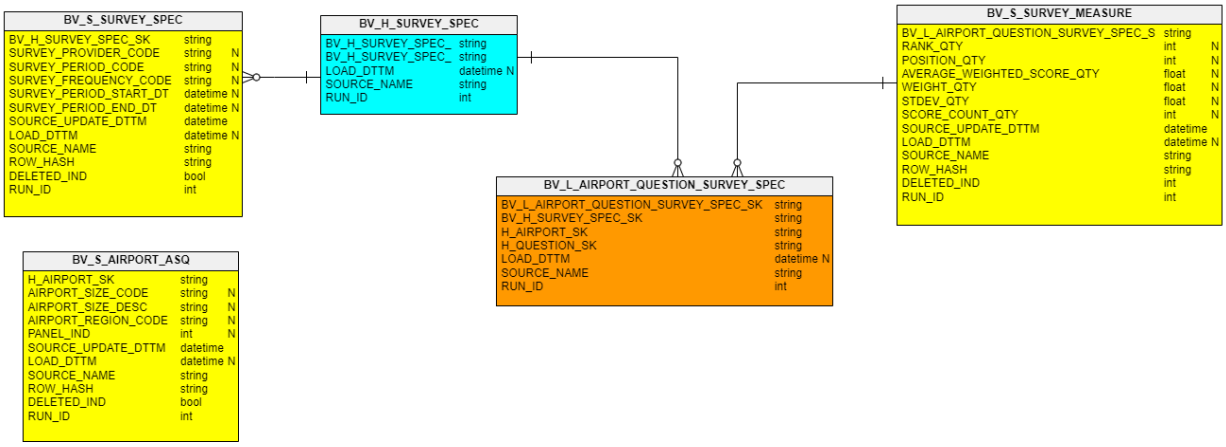
group	score	avg	avg per group	avg of avg
A	1	25.75	1	50.5
A	1			
A	1			
B	100		100	

This will result in the following business vault satellite:

BV_S_SURVEY_MEASURE		
BV_L_AIRPORT_QUESTION_SURVEY_SPEC_S	string	
RANK_QTY	int	N
POSITION_QTY	int	N
AVERAGE_WEIGHTED_SCORE_QTY	float	N
WEIGHT_QTY	float	N
STDEV_QTY	float	N
SCORE_COUNT_QTY	int	N
SOURCE_UPDATE_DTTM	datetime	
LOAD_DTTM	datetime	N
SOURCE_NAME	string	
ROW_HASH	string	
DELETED_IND	int	
RUN_ID	int	

7.1.5 The whole Business Vault

The new business vault tables will reside in the same schema as the data vault, some tables are even connected to existing data vault tables.



The final picture could look like this:

